Sl. No.	Author(s) & Year	Title of Study	Summary of paper	Research Focus / Context	Methodology / Approach	Key Findings	Research Gap / Limitation	Remarks	Narrative Version (for Paragraph Form in Literature Review Section)
1	Zhang & Li, 2022	A Review of Lightweight Design for Space Mirror Core Structure: Tradition and Future	Reviews the evolution of lightweight mirror cores from conventional contoured-back, open-back and sandwich forms to topology-optimized, lattice and Voronoi structures. It compares design strategies with emphasis on mass, stiffness and surface-figure preservation.	Lightweight structural design of space and telescope mirrors; development from conventional machining to additive-manufactured architectures.	Literature survey and cross-study comparison covering materials, lightweight ratios, structural forms and fabrication routes.	Shows that no single topology dominates across all requirements. The review identifies additive manufacturing and lattice/Voronoi concepts as important directions for increasing geometric freedom.	A review cannot establish performance through one controlled benchmark. Reported results also vary in geometry, material, loading and manufacturing assumptions, making direct quantitative comparison difficult.	Use: field foundation. DOI: 10.3390/machines10111066	Zhang and Li [1] trace mirror lightweighting from conventional contoured-back, open-back and sandwich cores to topology-optimized, lattice and Voronoi structures, comparing each route on mass, stiffness and surface-figure preservation. Their central conclusion is that no single topology dominates every requirement, while additive manufacturing widens the available geometric freedom. Because the reviewed results rely on different geometries, materials, loads and manufacturing assumptions, they cannot be compared directly. This motivates an automated method that treats mass reduction as one objective among several competing ones.
2	He et al., 2023	A Review of Surrogate-Assisted Evolutionary Algorithms for Expensive Optimization Problems	Provides a structured review of surrogate-assisted evolutionary algorithms for optimization problems where objective or constraint evaluations are computationally expensive. It discusses common surrogate models, search strategies and treatment of constrained and multiobjective cases.	Reducing the number of costly simulations or experiments needed during optimization.	Taxonomy and comparative review of surrogate-assisted evolutionary methods, including strategies for sampling, constraints and different objective settings.	The review shows that surrogate modelling can substantially reduce expensive evaluations, while model selection, infill sampling and constraint handling remain central design choices.	Performance is highly problem-dependent. The review is general rather than specific to opto-mechanical mirror design, so geometry representation and engineering validity need domain-specific treatment.	Use: optimization methodology. DOI: 10.1016/j.eswa.2022.119495	He et al. [2] review surrogate-assisted evolutionary algorithms for problems in which every objective or constraint evaluation is computationally expensive, covering common surrogate models, search strategies and the handling of constrained and multiobjective cases. They show that surrogates can greatly reduce the number of costly evaluations, although model selection, infill sampling and constraint handling remain critical choices. Since the review is not specific to opto-mechanical mirrors, geometry representation and engineering validity need separate treatment. In mirror design, surrogates can therefore act as an acceleration layer, with high-fidelity analysis reserved for final verification.
3	Kelmendi & Pappas, 2025	AI-Driven Optimization of Functional Feature Placement in Automotive CAD	Introduces a PointNet-based model coupled with Python CAD automation to predict functional feature locations on 3D automotive surfaces. The training labels are derived from FEA and physical-prototyping evidence rather than from geometry alone.	Learning useful design decisions from 3D geometry and transferring the prediction directly into a parametric CAD workflow.	Point-cloud learning with PointNet, followed by Python-driven CAD feature insertion and evaluation on eight test surfaces.	The prototype achieved a 75% success rate on six of eight test surfaces, demonstrating the feasibility of connecting learned geometric predictions with programmatic CAD.	The evaluation set is very small and the application is limited to dashboard feature placement. The study also notes generalization and richer contextual rules as unresolved issues.	Use: AI-CAD precedent. DOI: 10.3390/a18090553	Kelmendi and Pappas [3] couple a PointNet model with Python CAD automation to predict functional feature locations on 3D automotive surfaces, using labels derived from FEA and physical prototyping rather than geometry alone. The prototype achieved a 75% success rate on six of eight test surfaces, showing that learned geometric predictions can drive a programmatic CAD process. The very small test set and the narrow dashboard application, however, leave generalization and richer contextual rules unresolved. Engineering-focused work therefore needs broader testing and stronger physical constraints than this precedent provides.
4	Song et al., 2024	Balancing Objective Optimization and Constraint Satisfaction in Expensive Constrained Evolutionary Multiobjective Optimization	Addresses the difficulty of using inaccurate surrogate predictions for constraints in expensive multiobjective optimization. The method switches between unconstrained and constrained search modes and selectively builds constraint surrogates when they are useful.	Keeping surrogate-assisted search computationally efficient without allowing approximation errors to dominate feasibility decisions.	Kriging-assisted evolutionary optimization with adaptive search-mode switching and data selection for constraint models; tested on established benchmark suites.	The method improves the balance between convergence toward promising regions and reliable movement into feasible regions, reducing unnecessary reliance on constraint surrogates.	The validation is primarily benchmark-based rather than tied to a physical CAD/FEA workflow. Engineering constraints with strong nonlinear or discrete geometric effects are not specifically studied.	Use: constrained optimization. DOI: 10.1109/TEVC.2023.3300181	Song et al. [4] tackle the problem that inaccurate surrogate predictions of constraints can mislead expensive multiobjective optimization. Their Kriging-assisted method switches between unconstrained and constrained search modes and builds constraint surrogates only when they are useful, improving the balance between convergence toward promising regions and reliable movement into feasible ones. Validation relies on benchmark suites rather than a physical CAD/FEA workflow, and strongly nonlinear or discrete geometric constraints are not studied. The work still shows that feasibility must be managed explicitly, which is directly relevant to minimum rib thickness, allowable deformation and modal-frequency limits in mirrors.
5	Zhang et al., 2022	Bayesian Regularization Algorithm Based Recurrent Neural Network Method and NSGA-II for the Optimal Design of the Reflector	Combines a Bayesian-regularized recurrent neural network with NSGA-II to approximate key reflector responses and accelerate multiobjective structural optimization. The predicted outputs include surface deformation-related measures, modal frequency and mass.	Fast multiobjective optimization of opto-mechanical reflector parameters under competing mass, stiffness and optical requirements.	FEA-generated design samples are used to train BR-RNN surrogates; NSGA-II searches the learned response surfaces before selected designs are checked against the engineering model.	The reported prediction accuracies for the studied reflector indicators exceed about 99.6%, and the hybrid optimizer provides a faster alternative to evaluating every candidate directly with simulation.	The model is tied to a particular reflector parameterization and simulation dataset. External validation on different mirror geometries or independent measured data is not the main focus.	Use: mirror-surrogate baseline. DOI: 10.3390/machines10010063	Zhang et al. [5] combine a Bayesian-regularized recurrent neural network with NSGA-II to approximate reflector responses such as surface deformation, modal frequency and mass, using FEA-generated samples for training. The hybrid optimizer searches the learned response surfaces and checks selected designs against the engineering model, giving a faster alternative to simulating every candidate; reported prediction accuracy for the studied indicators exceeds about 99.6%. The model is tied to one reflector parameterization and simulation dataset, with little external validation. It is a close precedent that a new study can extend through stronger geometry transfer, multiple fidelity levels and independent validation.
6	He et al., 2023	ChatEDA: A Large Language Model Powered Autonomous Agent for EDA	Presents an LLM-controlled agent that plans tasks, generates scripts and executes electronic-design-automation tools to move through a multi-step chip-design workflow. The emphasis is on using the model as a controller while specialist software performs the actual engineering operations.	Natural-language control and autonomous orchestration of complex engineering software.	AutoMage-based agent, task planning, script generation and execution through EDA tools, with experiments across varied design requirements.	The study demonstrates that an LLM can coordinate a toolchain rather than only generate text, making structured tool use and workflow planning practical.	The domain is electronic design rather than mechanical CAD/FEA. The approach does not by itself guarantee physical validity, safe parameter selection or correct engineering assumptions.	Use: agentic orchestration. DOI: 10.1109/MLCAD58807.2023.10299852	He et al. [6] present ChatEDA, an LLM-controlled agent that plans tasks, generates scripts and runs electronic-design-automation tools to move through a multi-step chip-design workflow, with the model acting as a controller while specialist software does the engineering work. Experiments across varied design requirements show that an LLM can coordinate a toolchain rather than only generate text. The domain is electronic rather than mechanical CAD/FEA, and the approach does not guarantee physical validity or safe parameter selection. For mirror design, deterministic physics, CAD and FEA modules would remain authoritative while the agent handles planning and tool selection.

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7	Pan et al., 2025	Computationally Expensive High-Dimensional Multiobjective Optimization via Surrogate-Assisted Reformulation and Decomposition	Introduces a two-phase surrogate-assisted evolutionary strategy for high-dimensional expensive multiobjective problems. The first phase focuses on convergence through problem reformulation, while the second separates the decision space to recover solution diversity.	Controlling simulation cost and loss of diversity when optimization involves many coupled decision variables.	TP-SAEA combines surrogate-assisted reformulation, low-dimensional search, decomposition and re-evaluation of selected solutions; experiments span synthetic and real-world high-dimensional cases.	The method reports strong convergence and diversity using a limited number of real evaluations, including 509 evaluations on the reported benchmark settings.	The study is algorithm-centric and not tailored to CAD geometry, discrete topology choices or physics-based feasibility constraints encountered in optical mirror design.	Use: high-dimensional optimization. DOI: 10.1109/TEVC.2024.3380327	Pan et al. [7] propose a two-phase surrogate-assisted strategy (TP-SAEA) for high-dimensional expensive multiobjective problems: the first phase drives convergence through problem reformulation, and the second separates the decision space to recover solution diversity. Using a limited number of real evaluations, including 509 evaluations in the reported benchmark settings, the method shows strong convergence and diversity across synthetic and real-world cases. The study is algorithm-centred and not tailored to CAD geometry, discrete topology choices or physics-based feasibility constraints. Its staged principle becomes relevant once mirror designs grow beyond a handful of thickness and spacing parameters.
8	Huang et al., 2026	Data-driven sustainable design of automobile seats: integrating surrogate modeling and multi-objective optimization	Builds a data-driven design process around 800 expert and student scoring records, converting qualitative requirements into measurable design and sustainability objectives. Random Forest surrogate models are tuned with Bayesian optimization before NSGA-II searches the resulting design space.	Demonstrating how data-driven surrogates and multiobjective optimization can combine performance requirements that are difficult to evaluate with a single metric.	FAHP/SET-based requirement quantification, Random Forest surrogate modelling, Bayesian hyperparameter optimization, NSGA-II and SHAP-based interpretation.	The integrated BO-RF-NSGA-II framework provides a compact route from human requirements to Pareto-style design exploration and interpretable feature effects.	The data are subjective scoring records from one design context, and the authors note that engineering feasibility is not yet fully incorporated into the optimization loop.	Use: general data-driven MOO. DOI: 10.1038/s41598-026-59091-9	Huang et al. [8] build a data-driven design process on 800 expert and student scoring records, converting qualitative requirements into measurable design and sustainability objectives. Random Forest surrogates tuned by Bayesian optimization feed NSGA-II, and SHAP analysis interprets the feature effects, giving a compact route from human requirements to Pareto-style exploration. The data come from subjective scoring in a single design context, and engineering feasibility is not yet fully included in the optimization loop. Although outside optics, the study shows how surrogate modelling, optimization and interpretation can be combined in one pipeline, and its candour about data limitations is useful when defining a mirror-design methodology.
9	Bangera & Castro, 2025	Deflation constraints for global optimization of composite structures	Introduces deflation constraints that discourage an optimizer from repeatedly converging to solutions already found. The technique uses geometric exclusion functions so gradient-based and heuristic searches can explore additional optima.	Improving design-space coverage when several physically distinct good solutions may exist.	New deflation constraints based on hypersphere, hypercube and hypercuboid exclusion regions; demonstrated on structural optimization cases using gradient and heuristic methods.	The approach can expose multiple distinct optima instead of repeatedly returning the same solution basin, helping preserve diversity during repeated searches.	The examples concern composite structures rather than optical mirrors. Choosing exclusion regions for mixed continuous-discrete topology spaces would require additional study.	Use: diversity mechanism. DOI: 10.1016/j.compstruct.2025.118916	Bangera and Castro [9] introduce deflation constraints, based on hypersphere, hypercube and hypercuboid exclusion regions, that discourage gradient-based and heuristic optimizers from repeatedly converging to solutions already found. Demonstrated on structural optimization cases, the technique exposes several distinct optima and preserves diversity across repeated searches. The examples concern composite structures, and choosing exclusion regions for mixed continuous-discrete topology spaces would need further study. The work highlights that one converged design does not mean the design space has been explored; a mirror optimizer could use similar ideas to retain structurally different candidates for later CAD/FEA comparison.
10	Xu et al., 2022	Design of a 2 m Primary Mirror Assembly Considering Fatigue Characteristics	Studies a 2 m space optical mirror assembly with topology and size optimization of the mirror and rigid connectors, followed by fatigue-oriented design of flexible supports. Thermal-mechanical vibration and surface abrasion are included in the reliability assessment.	Coupling lightweight structural design with dynamic and fatigue reliability of the support system.	Topology/size optimization, finite-element modal and stress analysis, fatigue-life calculation and experimental vibration validation.	The design reached 5.6 nm surface error, 119.16 Hz first mode and an estimated fatigue life of about 21.79 million cycles; vibration and stress responses were experimentally checked.	The study is a detailed single-system investigation. It does not provide a general surrogate or automated CAD/FEA framework capable of transferring the learned design behaviour to new mirror configurations.	Use: reliability and validation. DOI: 10.3390/app122010326	Xu et al. [10] study a 2 m space optical mirror assembly through topology and size optimization of the mirror and its rigid connectors, followed by fatigue-oriented design of the flexible supports, with thermal-mechanical vibration and surface abrasion included in the reliability assessment. The final design reached 5.6 nm surface error, a 119.16 Hz first mode and an estimated fatigue life of about 21.79 million cycles, with vibration and stress responses checked experimentally. Being a detailed single-system study, it provides no general surrogate or automated CAD/FEA framework. It nevertheless shows that fatigue and vibration can decide practical viability, and its experimental comparison offers a benchmark for validating simulation.
11	Kim et al., 2023	Design optimization of a flexure spring used in small-sized ultra-precise optical instrument	Optimizes a compact Panto-style flexure spring using five design variables and a structured design-of-experiments approach. Response-surface modelling and KNN are compared with finite-element predictions and physical measurements.	Selecting support parameters that balance displacement, stress and manufacturability in precision optical mechanisms.	Taguchi L27 and face-centred CCD experiments, FEA, RSM regression and KNN prediction, followed by physical testing.	Thickness had the strongest effect among the studied variables, and combining RSM with KNN improved prediction compared with RSM alone for the tested responses.	The work focuses on a support spring rather than the mirror core and uses a small experimental design. Its learned relations are not demonstrated across different support geometries.	Use: support optimization / surrogate precedent. DOI: 10.1016/j.heliyon.2023.e22560	Kim et al. [11] optimize a compact Panto-style flexure spring with five design variables, using Taguchi L27 and face-centred CCD experiments, FEA, response-surface regression and KNN prediction, followed by physical testing. Thickness had the strongest effect among the studied variables, and combining RSM with KNN improved prediction over RSM alone for the tested responses. The work addresses a support spring rather than the mirror core and relies on a small experimental design, so its learned relations are not shown to hold across different support geometries. The study nevertheless demonstrates the value of pairing designed experiments with more than one regression model when support behaviour is nonlinear.
12	Lee et al., 2025	Design optimization of a lightweight secondary mirror assembly for a 1-m class ground telescope for satellite laser ranging	Presents a partially open-back hexagonal-cell secondary mirror and an optimized bipod flexure mount for a 1 m-class telescope. The design considers surface error in both vertical and horizontal pointing directions before fabrication and assembly.	Managing gravity-induced deformation and support errors in a lightweight secondary mirror.	Parametric lightweight-pocket optimization, flexure-mount optimization, FEA for multiple telescope orientations, fabrication and surface-quality testing.	The fabricated assembly demonstrates the practicality of integrating a lightweight mirror with an optimized flexure mount rather than optimizing the mirror core in isolation.	The study remains a dedicated design exercise for one telescope configuration and does not generalize its design process through a surrogate or autonomous optimization framework.	Use: support + fabrication validation. DOI: 10.1080/15599612.2025.2467152	Lee et al. [12] present a partially open-back hexagonal-cell secondary mirror and an optimized bipod flexure mount for a 1 m-class telescope, evaluating surface error in both vertical and horizontal pointing directions through parametric pocket and flexure optimization, FEA, fabrication and surface-quality testing. The fabricated assembly demonstrates the practicality of integrating a lightweight mirror with an optimized mount rather than optimizing the mirror core in isolation. It remains a dedicated design for a single telescope configuration, without a surrogate or autonomous framework. Because support configuration and gravity direction change the observed surface error, a surrogate should be trained on multiple load cases rather than one fixed boundary condition.

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13	Wang et al., 2024	Ensemble learning based hierarchical surrogate model for multi-fidelity information fusion	Develops EL-MFS, a hierarchical multi-fidelity surrogate that combines an ensemble of low-fidelity predictors with a smaller high-fidelity sample set. The design is intended to remain useful when the relationship between fidelity levels is not uniform.	Learning an accurate high-fidelity response without paying the cost of generating a large high-fidelity dataset.	Adaptive ensemble surrogate construction, feature mapping and hierarchical fusion of low- and high-fidelity information, tested on numerical and engineering examples.	The approach reduces sensitivity to a poorly chosen low-fidelity model and improves local prediction by allowing multiple surrogates to contribute according to their performance.	The method is general-purpose; the low- and high-fidelity relationship in optical mirror physics can be much more structured and may need geometry-aware inputs.	Use: multi-fidelity modelling. DOI: 10.1016/j.aei.2024.102535	Wang et al. [13] develop EL-MFS, a hierarchical multi-fidelity surrogate that combines an ensemble of low-fidelity predictors with a smaller set of high-fidelity samples, so that it stays useful even when the relationship between fidelity levels is not uniform. Tested on numerical and engineering examples, it reduces sensitivity to a poorly chosen low-fidelity model and improves local prediction by letting several surrogates contribute according to their performance. The method is general-purpose, whereas the low- and high-fidelity relationship in mirror physics is more structured and may need geometry-aware inputs. For mirrors, analytical mechanics could supply broad trends while selected NASTRAN runs correct the surrogate where geometry creates nonlinear effects.
14	Cheng et al., 2025	Fail-safe topology optimization for fiber-reinforced composite structures	Extends topology optimization to account for local damage by jointly considering structural layout and fibre-orientation variables. Damage is represented through local stiffness removal, and the optimization seeks designs that retain load-carrying performance under critical damage cases.	Embedding robustness and fault tolerance into structural optimization rather than checking damage only after the design is complete.	SIMP-based topology optimization with discrete-continuous fibre parameters, damage scenarios, adjoint sensitivities and a KS approximation for the max operator.	The method generates layouts that remain functional across prescribed local-damage scenarios and illustrates how damage-aware objectives can change the optimal structure.	The application is to composite structures, not optical mirrors. Translating fail-safe constraints to brittle glass-ceramic mirror ribs and optical surface tolerances requires domain-specific failure definitions.	Use: robust topology optimization. DOI: 10.1016/j.compstruct.2025.119145	Cheng et al. [14] extend topology optimization to local damage by jointly considering layout and fibre-orientation variables, representing damage as local stiffness removal and seeking designs that keep their load-carrying performance under critical damage cases. Using SIMP-based optimization with adjoint sensitivities and a KS approximation of the max operator, they obtain layouts that stay functional across prescribed damage scenarios and show how damage-aware objectives change the optimal structure. The application is to composite structures, so failure definitions for brittle glass-ceramic mirror ribs and optical tolerances would need domain-specific treatment. The robustness idea could still be valuable for thin mirror ribs once failure and optical-error criteria are defined.
15	Xu et al., 2023	Flexible support structure design for optical mirror	Develops a semi-kinematic flexible support for an aerial optical mirror and evaluates its behaviour under gravity, temperature change and assembly conditions. The design is analysed in FEA and then manufactured, assembled and measured.	Preserving optical surface accuracy by allowing controlled support compliance rather than imposing excessive constraints.	Kinematic analysis, flexible-support design, static/dynamic/thermal FEA, surface fitting, interferometric testing and vibration testing.	The measured assembly achieved l/52 RMS surface error and a 645 Hz fundamental frequency; simulation and dynamic test results were close, with about 4.2% frequency difference.	The design is specific to one support architecture and payload, and the study does not automate support selection across a broad design space.	Use: optical support validation. DOI: 10.1016/j.heliyon.2023.e20469	Xu et al. [15] develop a semi-kinematic flexible support for an aerial optical mirror and evaluate it under gravity, temperature change and assembly conditions through FEA, manufacture, assembly and measurement. The assembled mirror achieved λ/52 RMS surface error and a 645 Hz fundamental frequency, and simulation and dynamic tests agreed to within about 4.2% in frequency. The design is specific to one support architecture and payload and does not automate support selection across a broad design space. The study is a useful example of closing the loop between kinematic design, FEA and physical testing, and it reinforces the need to model support compliance and thermal freedom.
16	Zhu et al., 2023	Lightweight Design and Evaluation of Square Reflector	Investigates a 500 mm square secondary mirror made from ultra-low-expansion glass and introduces a gradually reinforced lightweight layout. The assembly is evaluated for mass, thermal-gravity surface error and modal behaviour.	Lightweighting of non-circular mirror geometries while maintaining optical stability.	Parametric structural design, finite-element simulation for gravity/temperature cases and experimental modal testing.	The reported mirror mass is 13.6 kg with 54.4 kg/m² surface density; measured first-order frequency is 167.8 Hz, with less than 3% difference from simulation.	The study focuses on a single square design and does not investigate automated topology discovery or data-driven transfer across mirror shapes.	Use: non-circular mirror baseline. DOI: 10.3390/app13169160	Zhu et al. [16] investigate a 500 mm square secondary mirror made of ultra-low-expansion glass, introducing a gradually reinforced lightweight layout evaluated for mass, thermal-gravity surface error and modal behaviour. The mirror mass is 13.6 kg (54.4 kg/m² surface density), and the measured first-order frequency of 167.8 Hz differs from simulation by under 3%. The study covers a single square design and does not investigate automated topology discovery or data-driven transfer across mirror shapes. It shows that the geometry class itself affects lightweighting and validation, so a learned design model restricted to circular mirrors may hide a meaningful generalization problem.
17	Zhang et al., 2025	OMS-DNN: A Surrogate Model for Forward Design of Mirror Structure and Optical System	Introduces a graph-enhanced deep surrogate model for forward mirror and optical-system design, with separate treatment of structural and optical variables. The work is built around large simulation datasets and uses optimization to recover parameter sets satisfying target constraints.	Replacing repeated FEA and ray tracing with learned forward models for rapid opto-mechanical design exploration.	Deep neural networks with graph convolution to encode parameter relationships, trained on about 20,000 structural and 100,000 optical-system samples, followed by optimization.	The reported model attains MAPE values of 0.98%, 4.09% and 1.66% on key outputs and supports rapid generation of constraint-compliant forward designs.	The data are simulation-generated and tied to the authors' parameterization; independent physical validation and broader cross-geometry transfer are not the central contribution.	Use: closest modern surrogate precedent. DOI: 10.1109/ACCESS.2025.3559481	Zhang et al. [17] introduce OMS-DNN, a graph-enhanced deep surrogate for forward design of mirror structures and optical systems that treats structural and optical variables separately. Trained on about 20,000 structural and 100,000 optical-system samples with graph convolution encoding parameter relationships, it reaches MAPE values of 0.98%, 4.09% and 1.66% on key outputs and supports rapid generation of constraint-compliant designs through optimization. The data are simulation-generated and tied to the authors' parameterization, with independent physical validation and cross-geometry transfer outside the main contribution. It leaves room to test whether geometry-aware and multi-fidelity inputs improve transfer to new mirror families.
18	Liu et al., 2023	Optimization Design of Large-Aperture Primary Mirror for a Space Remote Camera	Optimizes a 610 mm SiC primary mirror and its flexible hinge support using a compromise-programming formulation. The design combines structural modification, parametric optimization, finite-element analysis and experimental verification under gravity, temperature and assembly disturbances.	Integrated structural/support optimization for a large-aperture space-camera primary mirror.	Empirical initial sizing, parametric compromise programming, FEA of surface error and dynamics, followed by interferometric and vibration testing.	The final primary mirror mass is 8.66 kg; reported surface RMS is below l/50 under the stated combined conditions and the measured vibration frequency is 208.25 Hz.	The optimization remains tied to a specific mirror and material. Its workflow still depends on direct analysis and does not attempt surrogate-based transfer across a family of geometries.	Use: integrated mirror baseline. DOI: 10.3390/s23125441	Liu et al. [18] optimize a 610 mm SiC primary mirror and its flexible hinge support using compromise programming, combining structural modification, parametric optimization, FEA and experimental verification under gravity, temperature and assembly disturbances. The final mirror has a mass of 8.66 kg, a surface RMS below λ/50 under the stated combined conditions and a measured vibration frequency of 208.25 Hz. The optimization is tied to one mirror and material and still depends on direct analysis rather than surrogate-based transfer. It demonstrates that structural parameters, support layout and environmental conditions should be optimized together, and the verified design is a reference for judging whether an automated method gives credible solutions.

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19	Dong et al., 2025	Optimization of Ultra Lightweight Mirror and Opto-Mechanical-Thermal Coupling Analysis Based on Solar Thermal Radiation	Investigates an ultra-lightweight mirror through parameter sensitivity analysis, topology optimization and opto-mechanical-thermal evaluation under solar loading. The final structure is selected from several topology results and assessed using modal, optical and thermal metrics.	Balancing structural mass, stiffness and image quality when thermal radiation changes the mirror state.	MOAT and Sobol sensitivity analysis, multiobjective SIMP topology optimization, structural/modal FEA and ray-tracing-based optical evaluation under thermal conditions.	The optimized design shows a 30.37% weight reduction and 716.84 Hz first frequency; reported optical RMS improves by 8.95 nm while thermal-image quality remains close to the original case.	The process is computationally intensive and remains a one-system study. It does not introduce a surrogate or autonomous CAD loop to reuse the expensive opto-mechanical calculations.	Use: thermal-optical objective basis. DOI: 10.3390/s25020483	Dong et al. [19] investigate an ultra-lightweight mirror through MOAT and Sobol sensitivity analysis, multiobjective SIMP topology optimization and opto-mechanical-thermal evaluation under solar loading, selecting the final structure from several topology results. The optimized design shows a 30.37% weight reduction and a 716.84 Hz first frequency, with optical RMS improving by 8.95 nm while thermal-image quality stays close to the original case. The process is computationally intensive, covers one system, and offers no surrogate or autonomous CAD loop to reuse the expensive calculations. It shows that thermal distortion can change the optical ranking of a structural design, so thermal and optical responses belong in any surrogate-assisted optimization.
20	Zhang et al., 2022	Optomechanical Performances of Advanced Lightweight Mirrors Based on Additive Manufacturing	Compares an additively manufactured Voronoi lightweight mirror with a conventional honeycomb design. Both are assessed through an analysis flow that links lightweight structure, stiffness and optical-surface behaviour.	Evaluating whether AM-enabled cellular geometries can offer a better opto-mechanical trade-off than conventional machined cores.	Design-for-additive-manufacturing workflow, analytical honeycomb reference design and FEA comparison of the competing mirror architectures.	The Voronoi design performs better than the conventional control in area density, stiffness, surface stability and quilting deflection in the reported comparison.	The result is a topology comparison for selected designs rather than a general automated search over many geometries or manufacturing deviations.	Use: AM topology evidence. DOI: 10.3390/mi13081334	Zhang et al. [20] compare an additively manufactured Voronoi lightweight mirror with a conventional honeycomb design through a design-for-additive-manufacturing workflow, an analytical honeycomb reference and an FEA comparison. In the reported comparison the Voronoi design outperforms the control in area density, stiffness, surface stability and quilting deflection. The result concerns selected designs rather than a general automated search over many geometries or manufacturing deviations. It shows that alternative cellular geometries can alter several mirror metrics at once, strengthening the case for treating topology as a design choice instead of assuming one fixed pocket family.
21	Khan et al., 2025	Physics-informed geometric operators to support surrogate, dimension reduction and generative models for engineering design	Introduces geometry operators that augment low-level shape representations with descriptors derived from geometric and differential properties. The goal is to make surrogate and generative models more sensitive to shape characteristics that raw parameter values may miss.	Encoding geometry in a way that preserves physically meaningful shape information for data-driven engineering models.	Physics-informed geometric operators using quantities such as Fourier descriptors and curvature/integral information, evaluated for surrogate modelling, dimension reduction and generative tasks.	The study shows that enriching shape representations can improve the usefulness of data-driven models when shape characteristics carry information that simple coordinates do not capture.	The method is demonstrated on general engineering-shape problems rather than lightweight optical mirrors, so the most useful mirror-specific descriptors still need to be defined.	Use: geometry-aware ML. DOI: 10.1016/j.aei.2024.102937	Khan et al. [21] introduce physics-informed geometric operators that augment low-level shape representations with descriptors derived from geometric and differential properties, such as Fourier descriptors and curvature or integral information. Evaluated on surrogate modelling, dimension reduction and generative tasks, the enriched representations improve the usefulness of data-driven models when shape carries information that raw coordinates miss. The demonstrations use general engineering shapes rather than lightweight optical mirrors, so the most useful mirror-specific descriptors still need defining. For mirror cores this offers a route beyond plain scalar design vectors, since two parameter sets can produce structurally different cell layouts or curvature distributions.
22	Li et al., 2022	Structural Topology Optimization of Reflective Mirror Based on Objective of Wavefront Aberration	Optimizes reflective-mirror topology using wavefront aberration as the optical objective rather than relying only on structural displacement or mass. Deformed mirror surfaces are fitted with orthogonal bases and evaluated through ray tracing within a Cassegrain optical system.	Direct opto-mechanical optimization of imaging quality while controlling structural and mass-related constraints.	SIMP topology optimization, optical ray tracing, deformation fitting, adjoint sensitivity analysis and constrained multiobjective formulation.	The reported wavefront-aberration objective decreases substantially for the optimized primary and secondary mirrors, while the resulting structures satisfy the stated mass and rigid-body constraints.	The method still requires repeated high-fidelity structural and optical evaluations, and it is not embedded in a general-purpose automated CAD/surrogate framework.	Use: optical-objective foundation. DOI: 10.3390/machines10111043	Li et al. [22] optimize reflective-mirror topology using wavefront aberration as the optical objective instead of structural displacement or mass alone, fitting deformed surfaces with orthogonal bases and evaluating them by ray tracing in a Cassegrain system. Using SIMP, adjoint sensitivities and a constrained multiobjective formulation, they substantially reduce the wavefront-aberration objective for both primary and secondary mirrors while meeting the stated mass and rigid-body constraints. The method still needs repeated high-fidelity structural and optical evaluations and is not embedded in a general CAD/surrogate framework. The shift from structural proxies to an imaging-level objective is a strong basis for including wavefront error in a mirror optimizer.
23	Zhao & Li, 2023	The Application of a Topology Optimization Algorithm Based on the Kriging Surrogate Model in the Mirror Design and Optimization of an Aerial Camera	Builds a Kriging response surface for mirror RMS performance and embeds it inside a topology-optimization workflow. The surrogate is used to search for lightweight structures before the selected design is rechecked with finite-element analysis.	Reducing repeated FEA calls while optimizing mirror surface quality and lightweight structure.	Kriging surrogate modelling, Latin hypercube sampling, additive/multiobjective selection criteria and final FEM validation.	The reported optimized mirror reaches 10.41 nm RMS surface error, while the surrogate-to-FEA comparison shows an error of about 0.28%; the design also withstands the stated 20 g loading cases.	The study uses a single mirror configuration and one surrogate family. It does not explore transfer across mirror geometries, fidelity levels or independent experimental datasets.	Use: direct mirror-surrogate baseline. DOI: 10.3390/s23167236	Zhao and Li [23] build a Kriging response surface for mirror RMS performance and embed it in a topology-optimization workflow, using Latin hypercube sampling and final FEM validation of the selected design. The optimized mirror reaches 10.41 nm RMS surface error, the surrogate-to-FEA discrepancy is about 0.28%, and the design withstands the stated 20 g loading cases. The study uses a single mirror configuration and one surrogate family, without transfer across geometries, fidelity levels or independent experimental data. As one of the closest methodological precedents, it shows that a new study needs a clear extension beyond Kriging on one parameterization.
24	Wang et al., 2024	The Optimization Design of a Lightweight 2 m SiC Mirror for Ground-Based Telescopes	Develops a 2 m SiC primary mirror with a lotus-pattern back structure and optimizes structural variables after sensitivity screening. Major variables are explored with Latin-hypercube sampling and NSGA-II, followed by FEA and modal testing.	Large-aperture lightweight mirror design with coupled mass and surface-quality requirements.	Sensitivity analysis, Latin-hypercube design, NSGA-II, finite-element surface and modal analysis, and hammer-test validation.	The optimized mirror weighs 119 kg with 38.7 kg/m² areal density; FEA gives 11.3 nm surface RMS and 76.5 Hz first mode, with a 7.4% maximum relative frequency difference in the reported modal comparison.	The approach is computationally driven and system-specific; the optimizer still relies on a substantial direct simulation campaign and does not provide a reusable surrogate across mirror families.	Use: MOO + experimental baseline. DOI: 10.3390/photonics11060581	Wang et al. [24] develop a 2 m SiC primary mirror with a lotus-pattern back structure, screening variables by sensitivity analysis and exploring the major ones with Latin-hypercube sampling and NSGA-II, followed by FEA and hammer-test modal validation. The optimized mirror weighs 119 kg (38.7 kg/m² areal density), with 11.3 nm surface RMS and a 76.5 Hz first mode in FEA and a 7.4% maximum relative frequency difference in the reported modal comparison. The approach depends on a substantial direct simulation campaign and offers no reusable surrogate across mirror families. It is a practical baseline for judging whether a surrogate-assisted method can reach comparable designs with fewer costly simulations.

Sl. No.	Author(s) & Year	Title of Study	Summary of paper	Research Focus / Context	Methodology / Approach	Key Findings	Research Gap / Limitation	Remarks	Narrative Version (for Paragraph Form in Literature Review Section)
25	Jiang & Zhou, 2022	The Optimization of a Convex Aspheric Lightweight SiC Mirror and Its Optical Metrology	Designs a convex aspheric SiC secondary mirror together with an S-type flexure support, with surface accuracy considered under gravity, thermal and assembly effects. The final mirror is measured optically with a Hindle-based metrology setup and dynamically tested.	Preserving surface figure on a lightweight convex asphere under several disturbance sources.	Parameter optimization for mirror/support dimensions, finite-element structural evaluation, Hindle optical metrology and dynamic testing.	The measured surface accuracy reaches about l/100 RMS at the stated wavelength; the reported dynamic-test frequency is close to the finite-element prediction, with about 3.7% difference.	The study focuses on one convex-aspheric configuration and a dedicated metrology arrangement. It does not study data-driven generalization across topologies or automated multiobjective search.	Use: aspheric mirror validation. DOI: 10.3390/photonics9040210	Jiang and Zhou [25] design a convex aspheric SiC secondary mirror together with an S-type flexure support, considering surface accuracy under gravity, thermal and assembly effects and verifying the result with Hindle-based optical metrology and dynamic testing. The measured surface accuracy reaches about λ/100 RMS, and the dynamic-test frequency differs from the finite-element prediction by about 3.7%. The study concerns one convex-aspheric configuration with a dedicated metrology set-up and does not examine data-driven generalization across topologies or automated multiobjective search. It shows why the metrology path matters for aspheric, highly lightweighted mirrors, and models how independent physical evidence can test computational predictions.
26	Bourgenot et al., 2024	Topology Optimization of a Single-Point Diamond-Turning Fixture for a Deployable Primary Mirror Telescope	Applies topology optimization and additive manufacturing to a diamond-turning fixture for a deployable segmented telescope, targeting lower fixture mass and reduced machining deformation. The work evaluates conventional and optimized fixture concepts under relevant rotational and cutting loads.	Reducing manufacturing-tool mass without sacrificing the stiffness needed for ultra-precision optical machining.	Finite-element topology optimization, two mass-target cases, additive-manufacturing concepts and deformation comparison under machining loads.	The reported designs reduce mass by 66.45% and 34.19% in the two cases, with substantial reductions in deformation from rotational and cutting loads.	The study optimizes the fixture rather than the mirror, and the results are specific to a deployable machining setup. Its concepts are not integrated with a closed mirror-design optimization loop.	Use: manufacturing-process extension. DOI: 10.3390/aerospace11010050	Bourgenot et al. [26] apply topology optimization and additive manufacturing to a diamond-turning fixture for a deployable segmented telescope, aiming at lower fixture mass and less machining deformation under rotational and cutting loads. For the two mass-target cases the designs cut mass by 66.45% and 34.19%, with substantial reductions in deformation. The study optimizes the fixture rather than the mirror, and its concepts are specific to a deployable machining setup and not integrated with a closed mirror-design loop. It extends lightweighting to the tooling that determines the final shape, suggesting future mirror-design automation could include manufacturing constraints and fixture behaviour in the same workflow.

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